

ISCSI

In this SOP, I'm going to show how to install and configure the ISCSI target server and Initiator server.

ISCSI (Internet Small Computer Systems Interface) is a transport layer protocol that enables SCSI commands to be sent over standard TCP/IP networks. This technology allows you to create network-attached block storage devices, effectively turning any server with available storage into a SAN (Storage Area Network) without the need for expensive Fibre Channel infrastructure.

Role	Example
ISCSI Target (storage server)	192.168.163.129
ISCSI Initiator (client)	192.168.163.130

TARGET SERVER:-

- Install required package

```
nalien@localhost:~$ sudo dnf install -y targetcli
[sudo] password for nalien:
CentOS Stream 10 - BaseOS
CentOS Stream 10 - AppStream
CentOS Stream 10 - Extras packages
Extra Packages for Enterprise Linux 10 - x86_64
Extra Packages for Enterprise Linux 10 - x86_64
Package targetcli-2.1.58-5.el10.noarch is already installed.
Dependencies resolved.
Nothing to do.
Complete!
```

- Before doing the configuration, check the disk details.

```
nalien@localhost:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda          8:0    0   20G  0 disk
├─sda1       8:1    0    2M  0 part
├─sda2       8:2    0   953M  0 part /boot
├─sda3       8:3    0   14G  0 part /
└─sda4       8:4    0    1.9G  0 part [SWAP]
sdb          8:16   0    5G   0 disk
sr0         11:0    1   8.6G  0 rom
nalien@localhost:~$ df -hTP
Filesystem      Type      Size  Used Avail Use% Mounted on
/dev/sda3       xfs       14G   12G   2.6G  82% /
devtmpfs        devtmpfs  1.8G    0   1.8G   0% /dev
tmpfs           tmpfs     1.8G    0   1.8G   0% /dev/shm
tmpfs           tmpfs     725M   9.6M  715M   2% /run
tmpfs           tmpfs     1.0M    0   1.0M   0% /run/credentials/systemd-journald.service
/dev/sda2       xfs       889M  535M  355M  61% /boot
tmpfs           tmpfs     363M   72K  363M   1% /run/user/42
tmpfs           tmpfs     363M   56K  363M   1% /run/user/1001
```

- Check the target service status

```

nalien@localhost:~$ systemctl status --now target
● target.service - Restore LIO kernel target configuration
   Loaded: loaded (/usr/lib/systemd/system/target.service; enabled; preset: disabled)
   Active: active (exited) since Sat 2026-04-18 21:15:19 IST; 6s ago
  Invocation: 0de706e80ae2489faba573d8e7ab010d
    Process: 3378 ExecStart=/usr/bin/targetctl restore (code=exited, status=0/SUCCESS)
   Main PID: 3378 (code=exited, status=0/SUCCESS)
    Mem peak: 10.6M
      CPU: 86ms

Apr 18 21:15:19 localhost.localdomain systemd[1]: Starting target.service - Restore LIO kernel target configuration...
Apr 18 21:15:19 localhost.localdomain systemd[1]: Finished target.service - Restore LIO kernel target configuration.

```

- Once the disk details and service status are gathered, follow the steps below to proceed with the iSCSI configurations.
- Create storage (Backstore)

```

nalien@localhost:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda         8:0    0   20G  0 disk
├─sda1      8:1    0    2M  0 part
├─sda2      8:2    0   953M  0 part /boot
├─sda3      8:3    0   14G  0 part /
└─sda4      8:4    0   1.9G  0 part [SWAP]
sdb         8:16    0    5G  0 disk
sr0        11:0    1   8.6G  0 rom

nalien@localhost:~$
nalien@localhost:~$
nalien@localhost:~$ targetcli
opening lockfile failed: [Errno 13] Permission denied: '/var/run/targetcli.lock'
nalien@localhost:~$ sudo targetcli
targetcli shell version 2.1.58
Copyright 2011-2013 by Datera, Inc and others.
For help on commands, type 'help'.

/> ls
o- / ..... [Storage Objects: 0]
o- backstores ..... [Storage Objects: 0]
o- block ..... [Storage Objects: 0]
o- fileio ..... [Storage Objects: 0]
o- pscsi ..... [Storage Objects: 0]
o- ramdisk ..... [Storage Objects: 0]
o- iscsi ..... [Targets: 0]
o- loopback ..... [Targets: 0]
/> cd backstores/block
/backstores/block> ls
o- block ..... [Storage Objects: 0]
/backstores/block> /backstores/block create name=iscsi_disk dev=/dev/sdb
Created block storage object iscsi_disk using /dev/sdb.
/backstores/block> ls
o- block ..... [Storage Objects: 1]
o- iscsi_disk ..... [/dev/sdb (5.0GiB) write-thru deactivated]
o- alua ..... [ALUA Groups: 1]
o- default_tg_pt_gp ..... [ALUA state: Active/optimized]

```

- Create iSCSI Target (IQN)
- Configure LUN
- Allow access (ACL)
- Save configuration

```

/backstores/block> /iscsi create iqn.2026-04.lab.local:target1
Created target iqn.2026-04.lab.local:target1.
Created TPG 1.
Global pref auto_add_default_portal=true
Created default portal listening on all IPs (0.0.0.0), port 3260.
/backstores/block> cd /iscsi/
/iscsi> ls
o- iscsi ..... [Targets: 1]
  o- iqn.2026-04.lab.local:target1 ..... [TPGs: 1]
    o- tpg1 ..... [no-gen-acls, no-auth]
      o- acls ..... [ACLs: 0]
      o- luns ..... [LUNs: 0]
      o- portals ..... [Portals: 1]
        o- 0.0.0.0:3260 ..... [OK]
/iscsi> cd iqn.2026-04.lab.local:target1/
/iscsi/iqn.20...local:target1> ls
o- iqn.2026-04.lab.local:target1 ..... [TPGs: 1]
  o- tpg1 ..... [no-gen-acls, no-auth]
    o- acls ..... [ACLs: 0]
    o- luns ..... [LUNs: 0]
    o- portals ..... [Portals: 1]
      o- 0.0.0.0:3260 ..... [OK]
/iscsi/iqn.20...local:target1> cd
< @last tpg1/ path=
/iscsi/iqn.20...local:target1> cd tpg1/
tpg1/acls/ tpg1/luns/ tpg1/portals/
/iscsi/iqn.20...local:target1> cd tpg1/luns
/iscsi/iqn.20...et1/tpg1/luns> ls
o- luns ..... [LUNs: 0]
/iscsi/iqn.20...et1/tpg1/luns> create /backstores/block/iscsi_disk
Created LUN 0.
/iscsi/iqn.20...et1/tpg1/luns> ls
o- luns ..... [LUNs: 1]
  o- lun0 ..... [block/iscsi_disk (/dev/sdb) (default_tg_pt_gp)]
/iscsi/iqn.20...et1/tpg1/luns> cd ../acls
/iscsi/iqn.20...et1/tpg1/acls> ls
o- acls ..... [ACLs: 0]
/iscsi/iqn.20...et1/tpg1/acls> create iqn.2026-04.lab.local:client1
Created Node ACL for iqn.2026-04.lab.local:client1
Created mapped LUN 0.
/iscsi/iqn.20...et1/tpg1/acls> ls
o- acls ..... [ACLs: 1]
  o- iqn.2026-04.lab.local:client1 ..... [Mapped LUNs: 1]
    o- mapped_lun0 ..... [lun0 block/iscsi_disk (rw)]
/iscsi/iqn.20...et1/tpg1/acls> cd ../portals
/iscsi/iqn.20.../tpg1/portals> ls
o- portals ..... [Portals: 1]
  o- 0.0.0.0:3260 ..... [OK]

```

- Firewall

```

/iscsi/iqn.20.../tpg1/portals> exit
Global pref auto_save_conf=true
Last 10 configs saved in /etc/target/backup/.
Configuration saved to /etc/target/saveconf.json
nalien@localhost:~$ firewall-cmd --add-port=3260/tcp --permanent
firewall-cmd --reload
Authorization failed.
Make sure polkit agent is running or run the application as superuser.
Authorization failed.
nalien@localhost:~$ systemctl status firewalld.service
● firewalld.service - firewalld - dynamic firewall daemon
   Loaded: loaded (/usr/lib/systemd/system/firewalld.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-04-18 21:12:31 IST; 9min ago
   Invocation: 8a4729f832a50265b06178cdc2e6cbf
     Docs: man:firewalld(1)
   Main PID: 1131 (firewalld)
     Tasks: 2 (Limit: 22783)
   Memory: 58.3M (peak: 52M)
      CPU: 2.83s
   CGroup: /system.slice/firewalld.service
           └─1131 /usr/bin/python3 -sP /usr/sbin/firewalld --nofork --nopid

Apr 18 21:12:35 localhost.localdomain firewalld[1131]: WARNING: COMMAND_FAILED: '/usr/sbin/ip6tables -w10 -t filter -F DOCKER-CT' failed: ip6tables: No chain/target/match by that name.
Apr 18 21:12:35 localhost.localdomain firewalld[1131]: WARNING: COMMAND_FAILED: '/usr/sbin/ip6tables -w10 -t filter -X DOCKER-CT' failed: ip6tables: No chain/target/match by that name.
Apr 18 21:12:35 localhost.localdomain firewalld[1131]: WARNING: COMMAND_FAILED: '/usr/sbin/ip6tables -w10 -t filter -F DOCKER-INTERNAL' failed: ip6tables: No chain/target/match by that name.
Apr 18 21:12:35 localhost.localdomain firewalld[1131]: WARNING: COMMAND_FAILED: '/usr/sbin/ip6tables -w10 -t filter -X DOCKER-INTERNAL' failed: ip6tables: No chain/target/match by that name.
Apr 18 21:12:35 localhost.localdomain firewalld[1131]: WARNING: COMMAND_FAILED: '/usr/sbin/ip6tables -w10 -t filter -F DOCKER-ISOLATION-STAGE-1' failed: ip6tables: No chain/target/match by that name.
Apr 18 21:12:35 localhost.localdomain firewalld[1131]: WARNING: COMMAND_FAILED: '/usr/sbin/ip6tables -w10 -t filter -X DOCKER-ISOLATION-STAGE-1' failed: ip6tables: No chain/target/match by that name.
Apr 18 21:12:35 localhost.localdomain firewalld[1131]: WARNING: COMMAND_FAILED: '/usr/sbin/ip6tables -w10 -t filter -F DOCKER-ISOLATION-STAGE-2' failed: ip6tables: No chain/target/match by that name.
Apr 18 21:12:35 localhost.localdomain firewalld[1131]: WARNING: COMMAND_FAILED: '/usr/sbin/ip6tables -w10 -t filter -X DOCKER-ISOLATION-STAGE-2' failed: ip6tables: No chain/target/match by that name.
Apr 18 21:12:35 localhost.localdomain firewalld[1131]: WARNING: COMMAND_FAILED: '/usr/sbin/ip6tables -w10 -t filter -F DOCKER-ISOLATION' failed: ip6tables: No chain/target/match by that name.
Apr 18 21:12:35 localhost.localdomain firewalld[1131]: WARNING: COMMAND_FAILED: '/usr/sbin/ip6tables -w10 -t filter -X DOCKER-ISOLATION' failed: ip6tables: No chain/target/match by that name.
nalien@localhost:~$ sudo firewall-cmd --add-port=3260/tcp --permanent
[sudo] password for nalien:
success
nalien@localhost:~$ firewall-cmd --reload
Authorization failed.
Make sure polkit agent is running or run the application as superuser.
nalien@localhost:~$ sudo firewall-cmd --reload
success
nalien@localhost:~$ sudo firewall-cmd --list-ports
3260/tcp
nalien@localhost:~$ |

```

```

o- / ..... [...]
o- backstores ..... [...]
| o- block ..... [Storage Objects: 1]
| | o- iscsi_disk ..... [/dev/sdb (5.0GiB) write-thru activated]
| | o- alua ..... [ALUA Groups: 1]
| | o- default_tg_pt_gp ..... [ALUA state: Active/optimized]
| o- fileio ..... [Storage Objects: 0]
| o- pscsi ..... [Storage Objects: 0]
| o- ramdisk ..... [Storage Objects: 0]
o- iscsi ..... [Targets: 1]
| o- iqn.2026-04.lab.local:target1 ..... [TPGs: 1]
| | o- tpg1 ..... [no-gen-acls, no-auth]
| | o- acls ..... [ACLs: 1]
| | | o- iqn.2026-04.lab.local:client1 ..... [Mapped LUNs: 1]
| | | o- mapped_lun0 ..... [lun0 block/iscsi_disk (rw)]
| | o- luns ..... [LUNs: 1]
| | | o- lun0 ..... [block/iscsi_disk (/dev/sdb) (default_tg_pt_gp)]
| o- portals ..... [Portals: 1]
| | o- 0.0.0.0:3260 ..... [OK]
o- loopback ..... [Targets: 0]

```

CLIENT

- Install initiator tools
- Check the service status service
- Set Initiator Name
- Discover Target

```

root@localhost:~# dnf install -y iscsi-initiator-utils
Last metadata expiration check: 4:43:22 ago on Sat 18 Apr 2026 04:40:11 PM IST.
Package iscsi-initiator-utils-6.2.1.11-0.git4b3e853.el10.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
root@localhost:~# systemctl enable --now iscsid
Created symlink '/etc/systemd/system/multi-user.target.wants/iscsid.service' → '/usr/lib/systemd/system/iscsid.service'.
root@localhost:~# cat /etc/iscsi/initiatorname.iscsi
InitiatorName=iqn.1994-05.com.redhat:blid982816c0
root@localhost:~# vi /etc/iscsi/initiatorname.iscsi
root@localhost:~# cat /etc/iscsi/initiatorname.iscsi
InitiatorName=iqn.2026-04.lab.local:client1
root@localhost:~# systemctl restart iscsid
root@localhost:~# systemctl status iscsid
● iscsid.service - Open-iSCSI
   Loaded: loaded (/usr/lib/systemd/system/iscsid.service; enabled; preset: disabled)
   Active: active (running) since Sat 2026-04-18 21:26:23 IST; 6s ago
  Invocation: bd6a871fff36b48399fbce81d2f17145c
 TriggeredBy: ● iscsid.socket
    Docs: man:iscsid(8)
          man:iscsiuio(8)
          man:iscsiadm(8)
 Main PID: 34273 (iscsid)
   Status: "Ready to process requests"
    Tasks: 1 (Limit: 10320)
  Memory: 1.9M (peak: 2M)
     CPU: 8ms
    CGroup: /system.slice/iscsid.service
            └─34273 /usr/sbin/iscsid -f

Apr 18 21:26:23 localhost.localdomain systemd[1]: Starting iscsid.service - Open-iSCSI...
Apr 18 21:26:23 localhost.localdomain systemd[1]: Started iscsid.service - Open-iSCSI.
root@localhost:~# iscsiadm -m discovery -t sendtargets -p 192.168.163.129
192.168.163.129:3260,1 iqn.2026-04.lab.local:target1
root@localhost:~# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda           8:0    0   20G  0 disk
├─sda1        8:1    0    1M  0 part
├─sda2        8:2    0    1G  0 part /boot
├─sda3        8:3    0   19G  0 part
├─cs-root    253:0   0   17G  0 lvm  /
├─cs-swap    253:1   0    2G  0 lvm  [SWAP]
sr0          11:0    1   8.6G  0 rom

root@localhost:~# iscsiadm -m node -l
Login to [iface: default, target: iqn.2026-04.lab.local:target1, portal: 192.168.163.129,3260] successful.

```

- Login to Target
- Verify Disk
- Create filesystem
- Manual mounting
- Auto-mounting while starting or restarting the servers

```

root@localhost:~# iscsiadm -m node -l
Login to [iface: default, target: iqn.2026-04.lab.local:target1, portal: 192.168.163.129,3260] successful.
root@localhost:~# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda          8:0    0   20G  0 disk
├─sda1       8:1    0    1M  0 part
├─sda2       8:2    0    1G  0 part /boot
├─sda3       8:3    0   19G  0 part
│   └─cs-root 253:0    0   17G  0 lvm  /
│   └─cs-swap 253:1    0    2G  0 lvm  [SWAP]
sdb          8:16   0    5G  0 disk
sr0         11:0    1   8.6G  0 rom
root@localhost:~# mkfs.xfs /dev/sdb
meta-data=/dev/sdb          isize=512    agcount=4, agsize=327680 blks
                     =                  sectsz=512    attr=2, projid32bit=1
                     =                  crc=1        finobt=1, sparse=1, rmapbt=1
                     =                  reflink=1     bigtime=1 inobtcount=1 nrext64=1
                     =                  exchange=0    metadir=0
data        =                  bsize=4096    blocks=1310720, imaxpct=25
                     =                  sunit=0     swidth=0 blks
naming      =version 2        bsize=4096    ascii-ci=0, ftype=1, parent=0
log         =internal log     bsize=4096    blocks=16384, version=2
                     =                  sectsz=512    sunit=0 blks, lazy-count=1
realtime    =none            extsz=4096    blocks=0, rtextents=0
                     =                  rgcount=0    rgsz=0 extents
                     =                  zoned=0     start=0 reserved=0
root@localhost:~# mkdir /mnt/iscsi
root@localhost:~# mount /dev/sdc /mnt/iscsi
mount: /mnt/iscsi: fsconfig system call failed: /dev/sdc: Can't lookup blockdev.
dmesg(1) may have more information after failed mount system call.
root@localhost:~# mount /dev/sdb /mnt/iscsi
root@localhost:~# cat /etc/fstab | tail -1
/dev/vg_iscsi/lv_iscsiDisk1 /mnt/iscsiDisk1 xfs _netdev,nofail,x-systemd.automount,x-systemd.device-timeout=10 0 0
root@localhost:~# df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/mapper/cs-root 17G  4.3G   13G  25% /
devtmpfs        807M    0   807M   0% /dev
tmpfs           837M    0   837M   0% /dev/shm
tmpfs           335M    0   335M   0% /run
tmpfs           1.0M    0    1.0M   0% /run/credentials/systemd-journald.service
/dev/sda2       960M  423M   538M  44% /boot
tmpfs           168M    0   168M   0% /run/user/42
tmpfs           168M    0   168M   0% /run/user/0
/dev/sdb        5.0G  130M   4.9G   3% /mnt/iscsi
root@localhost:~# mount | grep iscsi
systemd-1 on /mnt/iscsiDisk1 type autofs (rw,relatime,fd=57,pgrp=1,timeout=0,minproto=5,maxproto=5,direct,pipe_ino=11683)
/dev/sdb on /mnt/iscsi type xfs (rw,relatime,seclabel,attr2,inode64,logbufs=8,logbsize=32k,noquota)
root@localhost:~#

```